Leeds-Harvard Referencing Style for BibLaTeX

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Abstract

This document demonstrates the `leeds-harvard` BibLaTeX style across authentic publications in Physics, Computer Science, and Engineering. It details the exact LaTeX commands, rendered in-text citations, reference list formatting, and specific rules in the Leeds-Harvard style.

1 In-Text Citations: Syntax & Output

Leeds-Harvard is an author–date referencing system. Table 1 details the standard citation commands alongside their rendered output:

Table 1: In-text citation syntax and rendered Leeds-Harvard output.

Citation Scenario	**LaTeX Command**	**Rendered Output**
One author (parenthetical)	`\parencite{shannon1948}`	(Shannon, 1948)
One author (narrative)	`\textcite{shannon1948}`	Shannon (1948)
Two authors	`\parencite{gencturk2015}`	(Gencturk and Hosseini, 2015)
Three or more authors	`\parencite{macnaughton1999}`	(MacNaughton et al., 1999)
Three or more (narrative)	`\textcite{clayden2012}`	Clayden et al. (2012)
Specific page	`\parencite[24]{shannon1948}`	(Shannon, 1948, p.24)
Page range	`\parencite[20-29]{shannon1948}`	(Shannon, 1948, pp.20-29)
Roman numerals	`\parencite[iv]{dirac1958}`	(Dirac, 1958, iv)
Unpaginated work	`\parencite[no pagination]{dirac1958}`	(Dirac, 1958, no pagination)
Same author, multiple years	`\parencite{knuth1968, knuth1984texbook,knuth1997}`	(Knuth, 1968; 1984; 1997)
Same author, same year	`\parencite{turing1937a, turing1937b}`	(Turing, 1937a; Turing, 1937b)
Same author, same year (narrative)	`\textcite{turing1937b}`	Turing (1937b)
Undated work (no date)	`\parencite{turingnd}`	(Turing, no date)
Undated work (narrative)	`\textcite{turingnd}`	Turing (no date)
Corporate author (first cite)	`\parencite{nasa2011}`	(National Aeronautics and Space Administration [NASA], 2011)
Corporate author (subsequent)	`\parencite{nasa2011}`	(NASA, 2011)
Secondary citation	`\parencite[Shannon, 1948, cited in][9]{tanenbaum2003}`	(Shannon, 1948, cited in Tanenbaum, 2003, p.9)

Citation Scenario	LaTeX Command	Rendered Output
Multiple works, different authors	`\parencite{macnaughton1999, ahmed2012,knuth1997}`	(MacNaughton et al., 1999; Ahmed and Meehan, 2012; Knuth, 1997)
Narrative list of authors	`\textcite{adams1906, strogatz1994,han2026}`	Adams (1906), Strogatz (1994) and Han (2026)
Anonymous work	`\parencite{anon2008}`	(Anon, 2008)
Forthcoming article	`\parencite{royallnd}`	(Royall et al., forthcoming)

2 FAQ: What Readers and Users Should Expect

Q: Why does Leeds-Harvard use the English word "and" instead of an ampersand ("&")?

What to expect: In Leeds-Harvard, multiple authors are **always** connected with the English word "and", both in in-text citations (e.g. (Gencturk and Hosseini, 2015)) and in the reference list (e.g. *Tanenbaum, A.S. and Wetherall, D.J.*).

- **Parenthetical citations:** Always write "and" ((Gencturk and Hosseini, 2015)), never an ampersand `(Gencturk & Hosseini, 2015)`.
- **Reference list:** Authors are joined by "and" without a serial (Oxford) comma: e.g. *Gencturk, B. and Hosseini, F.* and *MacNaughton, S.J., Stephen, J.R., ... and White, D.C.* (never `&`).

Q: How does Leeds-Harvard handle multiple publications by the same author?

What to expect: Leeds-Harvard compresses author names in parenthetical citations. If you cite multiple works by Donald Knuth, the author's surname is printed once followed by the publication years separated by semicolons:

- `\parencite{knuth1968,knuth1984texbook,knuth1997}`
  yields (Knuth, 1968; 1984; 1997).

Q: In what order should multiple citations appear inside the same parentheses?

What to expect: Leeds-Harvard specifies that multiple sources in one citation must be arranged **chronologically** (oldest first), separated by semicolons:

- (MacNaughton et al., 1999; Ahmed and Meehan, 2012; Knuth, 1997).
- **Important:** Because `sortcites=false` is set to honour the author's intentional citation order, list your BibTeX keys in chronological order in your `\parencite` call.

Q: When is "et al." used in citations versus the reference list?

What to expect:

- **In-Text Citations:**

  - 1 or 2 authors: List both surnames ((Shannon, 1948), (Gencturk and Hosseini, 2015)).
  - 3 or more authors: List only the first author followed by "et al." ((MacNaughton et al., 1999)).
  - Note that "et al." is written in upright roman font (**not** italicised) in Leeds-Harvard.

- **Reference List:** *All* authors must be explicitly named regardless of how many there are. The abbreviation "et al." is **never** permitted in the bibliography itself.

Q: How are page numbers, page ranges, roman numerals, and unpaginated works formatted?

What to expect:

- **Single page:** Use lowercase `p.` without a space: (Shannon, 1948, p.24).
- **Page ranges:** Use `pp.` with a hyphen and no space: (Shannon, 1948) (in bibliography: *pp.379-423*).
- **Roman numerals:** Do *not* add `p.` to roman numerals: `\parencite[iv]{dirac1958}` → (Dirac, 1958, iv).
- **No pagination:** If a work is unpaginated (e.g. an online article), specify `no pagination` in the postnote: `\parencite[no pagination]{dirac1958}` → (Dirac, 1958, no pagination).

Q: Why do corporate authors require double braces in BibTeX?

What to expect: If a corporate name is entered with single braces:

```
author = {National Aeronautics and Space Administration}
```

BibTeX parses "Administration" as the surname and "National Aeronautics and Space" as first names!

- **Double braces in `.bib`:** Corporate names must be protected with double curly braces:

```
author      = {{National Aeronautics and Space Administration}},
shortauthor = {NASA},
```

- **Automatic acronym expansion:** The first citation prints the full corporate name with the acronym in square brackets: (NASA, 2011). Subsequent citations automatically use the acronym: (NASA, 2011).

Q: What happens when a book lacks a place of publication or publisher?

What to expect:

- In standard Leeds-Harvard guidelines, works with an omitted place of publication historically printed `[no place]`.

- In modern academic practice, place of publication is often omitted for major publishers (e.g. (Hawking, 1988)). By default, `leeds-harvard` sets `no-place=false` to cleanly omit `[no place]`.
- To enforce strict Leeds-Harvard behaviour (where every missing publication place prints `[no place]: Publisher`), load the package with:

```
\usepackage[style=leeds-harvard, no-place=true]{biblatex}
```

- If a book has no publisher specified, it correctly prints `[no publisher]` (e.g. Gibbs 1881).

Q: Why should titles in the `.bib` file be in sentence case?

What to expect: Leeds-Harvard requires article and book titles to appear in **sentence case** (only the first word of the title, subtitle, and proper nouns capitalized). BibLaTeX preserves title capitalization as entered in your `.bib` file. Always protect proper nouns, acronyms, and formulas with curly braces:

```
title = {The {Byzantine} generals problem},
title = {Evaluation of reinforced concrete and reinforced
         engineered cementitious composite ({ECC}) members},
```

Q: Why is there no full stop at the end of reference entries ending in a URL or DOI?

What to expect: To avoid broken hyperlinks when readers copy text from a PDF, Leeds-Harvard strictly prohibits placing a full stop immediately after a URL or DOI. Trailing punctuation is automatically stripped by the package.

3 Physics-Specific Citations and Journals

In Physics and Astronomy, referencing conventions often rely on automated bibliographic exports (e.g. SAO/NASA ADS, arXiv, APS, AIP, IOP). Leeds-Harvard accommodates these workflows while maintaining university standards:

3.1 1. Full Journal Titles vs Standard Physics Macros

Physics authors routinely use standard AAS/APS macros in BibTeX exports (such as `\apjl`, `\mnras`, `\aap`, `\prl`, `\pra`). However, **Leeds-Harvard strictly forbids journal abbreviations**.

The `leeds-harvard` style includes an internal source map that automatically expands these macros into their full official titles during compilation:

- `\apjl` → *The Astrophysical Journal Letters*
- `\mnras` → *Monthly Notices of the Royal Astronomical Society*
- `\aap` → *Astronomy and Astrophysics*

- `\prl` → *Physical Review Letters*
- `\pra`, `\prb`, `\prc`, `\prd`, `\pre` → *Physical Review A*, *B*, *C*, *D*, *E*

You do **not** need to load `aas_macros.sty` or manually rewrite your BibTeX database.

3.2 2. Electronic Article Identifiers (`eid`)

Modern physics journals (APS *Physical Review*, AIP, IOP, *ApJ Letters*) use article identifiers instead of sequential pagination:

- (Kurikawa et al., 2020): *Physical Review Research*, **2**(2), article no: 023307 [no pagination].
- (Chen et al., 2026): *The Astrophysical Journal Letters*, **1008**(2), article no: L60 [no pagination].

The package detects `eid` (or `eid = pages` from ADS) and formats it as *article no: ... [no pagination].*

3.3 3. arXiv Preprints

Preprints from arXiv are fully supported with official Leeds-Harvard preprint formatting:

- Grachev et al. (2015) formats as:
  Grachev, A.A., Leo, L.S., Di Sabatino, S., Fernando, H.J.S., Pardyjak, E.R. and Fairall, C.W. 2015. [Pre-print]. Structure of turbulence in katabatic flows below and above the wind-speed maximum. arXiv. [Online]. Available from: `https://doi.org/10.48550/arXiv.1501.02297`

3.4 4. Classic Physics Treatises & Foundations

The style cleanly formats landmark physics monographs and papers:

- **Quantum Mechanics:** Dirac (1958, (Dirac, 1958)), Wheeler and Zurek (1983, (Wheeler and Zurek, 1983)), Bohm (1983, (Bohm, 1983)), Planck (1922, (Planck, 1922)).
- **Relativity & Classical Foundations:** Einstein (1920, (Einstein, 1920)), Landau and Lifshitz (1976, (Landau and Lifshitz, 1976)), Newton (1999, (Newton, 1999)), Lorentz (1952, (Lorentz, 1952)).
- **Astrophysics & Gravitation:** Lynden-Bell (1967, (Lynden-Bell, 1967)), Shakura and Sunyaev (1973, (Shakura and Sunyaev, 1973)), Sunyaev and Zeldovich (1970, (Sunyaev and Zeldovich, 1970)), Sample (2016, (Sample, 2016)).
- **Applied & Materials Physics:** Sallo and Olsen (1961, (Sallo and Olsen, 1961)), Royall et al. (2013, (Royall et al., 2013)).

4 Minimal Working Example (MWE)

Here is a minimal template for your own thesis, paper, or report:

```
\documentclass[a4paper,11pt]{article}
\usepackage[british]{babel}
\usepackage{csquotes}
\usepackage[style=leeds-harvard]{biblatex}
\addbibresource{references.bib}
\usepackage[hidelinks]{hyperref}

\begin{document}

According to \textcite{shannon1948}, entropy measures uncertainty.
This theorem was verified across networks \parencite{tanenbaum2011}.

\printbibliography % Automatically produces section titled "References"

\end{document}
```

The above MWE produces the following in-text output:

According to Shannon (1948), entropy measures uncertainty. This theorem was verified across networks (Tanenbaum and Wetherall, 2011).

And generates the following reference list entries:

> **References**
>
> Shannon, C.E. 1948. A mathematical theory of communication. *Bell System Technical Journal*. [Online]. **27**(3), pp.379-423. [Accessed 4 September 2024]. Available from: `https://doi.org/10.1002/j.1538-7305.1948.tb01338.x`
>
> Tanenbaum, A.S. and Wetherall, D.J. 2011. *Computer networks*. 5th ed. Boston: Pearson Prentice Hall.

Below is the complete bibliography of all works cited across this test document:

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